Fourier endpoint minimax and autocorrelation deficit rank bounds

A two theorem working manuscript

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Abstract

This manuscript gives complete proofs of two previously established project results, keeping their hypotheses and conclusions separate. For the cyclic Fourier matrix of order $N = 2^m \geq 8$, the minimum, over independent row and column permutations, of the maximum strict off-diagonal corner rank is $N/4 + 1$. Double bit reversal attains the minimum. The minimum rank of an arbitrary half-size square is instead $N/4$. Separately, for a unitary character table of any finite abelian group, a complementary Gram-matrix second moment gives a rectangular rank bound and a positive low-rank approximation gap in terms of the maximum nonzero-translation overlap of the column set. All degenerate sizes, normalization choices, and singular-value indices are specified below.

This is a consolidation of accepted proof sources, not a new theorem or fusion. Digit reversal, Fourier support uncertainty, the flat-unitary rank bound, translation-overlap statistics, and spectral approximation arguments have established antecedents. The exact scope's historical priority remains unresolved. No general FFT arithmetic-complexity conclusion or publication-acceptance claim is made.

1 Scope and conventions

For a finite set E, write $\mathbb{C}^E$ for its coordinate space with the standard Hermitian inner product. If A has row set E and column set D, then $A[R, H]$ is the submatrix with $R \subseteq E$ and $H \subseteq D$. Reordering the chosen rows or columns does not change rank or singular values. Complements are taken in the full relevant indexing set. An empty matrix has rank zero.

For $N \geq 1$, put $[N] = \{0, \ldots, N-1\}$. If ρ is a permutation of $[N]$, use the convention $(P_\rho x)_i = x_{\rho(i)}$. Thus

$$(P_\rho A P_\sigma^T)_{ij} = A_{\rho(i),\sigma(j)}.$$

For an $N \times N$ matrix define

$$q(A) = \max_{1 \leq k < N} \max\{\operatorname{rank} A[0:k, k:N],\ \operatorname{rank} A[k:N, 0:k]\}. \tag{1}$$

The interval notation is half-open. These two blocks are the strict off-diagonal corners across the same coordinate cut. Set $q(A) = 0$ for $N = 1$, when there is no cut. The two endpoint permutations in the minimization below are arbitrary and independent.

The unnormalized cyclic Fourier matrix and its unitary normalization are

$$F_N = (\omega_N^{ab})_{a,b\in[N]}, \qquad \omega_N = e^{2\pi i/N}, \qquad U_N = N^{-1/2}F_N. \tag{2}$$

For a finite abelian group G of order N, written additively, its dual group is $\widehat{G}$. The unitary character table used here is

$$U_{\chi,x} = N^{-1/2}\chi(x), \qquad \chi \in \widehat{G},\ x \in G. \tag{3}$$

Characters have modulus one. Character orthogonality makes U unitary. Conjugating the Fourier convention changes none of the conclusions. Nonzero scalar normalization does not change exact rank; it does change singular values and operator-norm errors. Throughout the analytic statements, $\|\cdot\|_2$ denotes the operator norm for Euclidean norms, rather than the Frobenius norm.

Exact assumption and convention table

Item	Convention and required hypothesis
Endpoint minimax	Cyclic $N = 2^m$; independent endpoint permutations; one matrix and a common cut coordinate order
Exact field scope in Theorem 1	Complex numbers, or a characteristic-zero field containing a primitive Nth root; no square root of N is needed for the unnormalized statement
Deficit theorem	Any finite abelian G, $N \geq 2$; complex unitary-normalized character table; $1 \leq r, h \leq N$
Selected sets	Rows $R \subseteq \widehat{G}$, columns $H \subseteq G$; $r = \lvert R\rvert$, $h = \lvert H\rvert$
Translation overlap	$c_H(s) = \lvert H \cap (H - s)\rvert = \lvert\{x \in H : x + s \in H\}\rvert$
Deficit	$d_H = h - \max_{s\neq 0} c_H(s)$; defined only when $N \geq 2$
Singular values	Descending square roots of the h eigenvalues of V^*V, including zeros if $h > r$
Approximation rank	$W \in \mathbb{C}^{r\times h}$, integer $t \geq 0$, $\operatorname{rank} W \leq t$; strict gap condition $t < h - B_H$
Half-square result	Cyclic $N = 4L$; arbitrary row and column sets, each of size $N/2$; $N = 2$ treated separately
Finite characteristic	No general lower-bound or positive-gap transfer is asserted; modular checks are controls only
Open extensions	Noncyclic matching minimax upper bounds and exact all-size minimum rank sums are not assumptions of either theorem

2 The cyclic dyadic endpoint theorem

Theorem 1. Let $N = 2^m$, $m \geq 0$. Over $\mathbb{C}$,

$$\min_{\rho,\sigma\in S_N} q(P_\rho F_N P_\sigma^T) = \begin{cases} 0, & N = 1, \\ 1, & N = 2 \text{ or } 4, \\ N/4 + 1, & N \geq 8. \end{cases} \tag{4}$$

For every size, bit reversal at both endpoints is a minimizing pair. The same exact statement holds over every characteristic-zero field containing a primitive Nth root, with that root in place of ω_N.

Separately, for $N = 2^m \geq 4$,

$$\min_{|R|=|H|=N/2} \operatorname{rank} F_N[R,H] = N/4. \tag{5}$$

For $N = 2$, this half-square minimum is 1. At $N = 1$ there is no integer half-square size. Equation (5) is not a $+1$ statement.

2.1 The universal critical rectangle lower bound

Write $N = 4L$ with $L \geq 2$. Take arbitrary $R, H \subseteq \mathbb{Z}_N$ with

$$|R| = 2L - 1, \qquad |H| = 2L + 1 =: h.$$

Let $I = \mathbb{Z}_N \setminus R$, so $|I| = h$, and let

$$V = U_N[R,H], \qquad A = U_N[I,H]^* U_N[I,H].$$

The columns of U_N are orthonormal, hence

$$0 \leq A \leq I_h, \qquad V^*V = I_h - A. \tag{6}$$

Consequently $\ker V$ is the eigenvalue-one eigenspace of A. Its dimension d is at most $\operatorname{tr}(A^2)$, since every eigenvalue of A belongs to $[0,1]$.

Set $f_I(s) = \sum_{a \in I} \omega_N^{as}$. The (x,y) entry of A, for $x, y \in H$, is $N^{-1} f_I(y-x)$. Grouping ordered pairs by their difference gives

$$\operatorname{tr}(A^2) = \frac{1}{N^2} \sum_{s \in \mathbb{Z}_N} c_H(s) |f_I(s)|^2. \tag{7}$$

Here $c_H(0) = h$. Every nonzero translation of $\mathbb{Z}_{2^m}$ has even order and partitions the group into cycles of that even length. An invariant set is a union of those cycles and therefore has even cardinality. Since h is odd, H is not invariant under any such translation. Thus $c_H(s) \leq h - 1$ for $s \neq 0$. Orthogonality also gives

$$\sum_s |f_I(s)|^2 = Nh, \qquad f_I(0) = h.$$

Substitution into (7) yields

$$\begin{aligned} \operatorname{tr}(A^2) &\leq \frac{hh^2 + (h-1)(Nh - h^2)}{N^2} \\ &= L + \frac{3}{4} + \frac{1}{4L} + \frac{1}{16L^2} =: B_L < L + 1. \end{aligned} \tag{8}$$

For the last inequality, $4L^2 - 4L - 1 > 0$ for every integer $L \geq 2$. Since d is an integer, $d \leq L$. Therefore every such rectangle satisfies

$$\operatorname{rank} F_N[R,H] = h - d \geq L + 1. \tag{9}$$

After any independent row and column reorderings, the upper-right corner at $k = 2L - 1$ has exactly these dimensions. Equation (9) therefore proves the lower bound in (4) uniformly over both permutations. No assumption about the positions of either selected set was used.

2.2 The exact bit reversal recurrence

Let $b_m(j)$ be reversal of the m binary digits of j, with $b_0(0) = 0$. Define

$$Q_N[i, j] = \omega_N^{b_m(i)b_m(j)}, \qquad r_N(k) = \operatorname{rank} Q_N[0 : k, k : N].$$

Put $r_N(0) = r_N(N) = 0$. We prove, for every dyadic $N = 4L \geq 4$ and $0 \leq k \leq 2L$,

$$r_N(k) = \lceil k/2 \rceil + r_L(\lfloor k/2 \rfloor). \tag{10}$$

The smaller table uses $\omega_L = \omega_N^4$, including the one-entry table when $L = 1$. The first $2L$ bit-reversed labels are all the even labels. Write them as $2q_i$. If $z_j = b_{m-2}(j)$, then

$$q_{2j} = z_j, \qquad q_{2j+1} = L + z_j, \qquad 0 \leq j < L. \tag{11}$$

Let $p = \lfloor k/2 \rfloor$ and $d = \lceil k/2 \rceil$. The selected rows contain p complete pairs from (11), and one unpaired row if k is odd. They have d distinct residues modulo L.

All odd natural column labels $2\ell + 1$, $0 \leq \ell < 2L$, lie to the right of the cut. In row label $2q$ their entries are

$$\omega_N^{(2q)(2\ell+1)} = \omega_{2L}^q \omega_L^{q\ell}. \tag{12}$$

Rows q and $q + L$ have opposite values on these columns. On one representative of each distinct residue, removal of the nonzero phase ω_{2L}^q leaves d distinct rows of the full Fourier matrix F_L, with its columns repeated twice. These rows are independent, so the odd-column span has dimension exactly d.

Define $\pi : \mathbb{C}^k \to \mathbb{C}^p$ by adding the two coordinates in each complete pair and ignoring the unpaired coordinate, if present. The odd-column span lies in $\ker \pi$. Since π is surjective and $k - p = d$, both spaces have dimension d and hence are equal. Passing to the quotient by the odd-column span is therefore precisely passing through π.

A surviving even column label 2ℓ has equal entries on each paired row, and its image under π is

$$2(\omega_L^{z_j \ell})_{0 \leq j < p}. \tag{13}$$

The column positions $k, \ldots, 2L - 1$ in (11) give, modulo L, exactly the residues $z_p, \ldots, z_{L-1}$, with possible repetitions. At an odd cut $k = 2p + 1$, the second member of the partially removed pair still survives and supplies z_p. Dropping it would change the recurrence incorrectly. The quotient columns therefore have rank $r_L(p)$. Adding back the d-dimensional odd-column span proves (10), including $p = 0$ and $p = L$.

2.3 Reflection and the maximum cut rank

Digit reversal satisfies $b_m(N-1-i) = N-1-b_m(i)$. Consequently

$$Q_N[N-1-i, N-1-j] = \omega_N^{1+b_m(i)+b_m(j)} Q_N[i,j]. \tag{14}$$

The factors on the right are invertible row and column scalings. Simultaneous reversal carries an upper corner at k to a lower corner at $N-k$; Q_N is symmetric, so

$$r_N(k) = r_N(N-k). \tag{15}$$

For $k = 2a \le 2L$, (10) and the column-count bound $r_L(a) \le L-a$ give $r_N(k) \le L$. For $k = 2a+1 < 2L$, they give $r_N(k) \le L+1$. Reflection covers every other cut. If $L \ge 2$, then $r_L(L-1) = 1$, because this is a nonempty single-column Fourier block with nonzero entries. Thus $r_N(2L-1) = L+1$. Symmetry equates the upper and lower ranks at each cut. This proves the matching upper bound for $N \ge 8$.

The bases are $r_1(0) = r_1(1) = 0$ and $r_2(1) = 1$. At $N = 4$, (10) and (15) give the profile $(1,1,1)$. Every nontrivial Fourier matrix has a nonzero corner at $k = 1$, so the minima at $N = 2, 4$ are exactly one. This also explains why substituting $N = 4$ into $N/4+1$ is invalid.

2.4 Half squares and the characteristic zero extension

For every complex Fourier rectangle $V = U_N[R,H]$, coordinate restriction gives $\|V\|_2 \le 1$, while $\|V\|_F^2 = |R||H|/N$. The sum of the squared singular values is at most their number, hence

$$\operatorname{rank} V \ge |R||H|/N. \tag{16}$$

For $N = 4L$ and half-size sets, this is $\operatorname{rank} V \ge L$. Choose both sets to be all even labels. With $0 \le a, b < 2L$, the unnormalized entries are

$$\omega_N^{(2a)(2b)} = \omega_L^{ab}.$$

Every residue modulo L occurs twice in each axis, so the block is F_L with repeated rows and columns and has rank exactly L. This proves (5). The chosen set has a nontrivial translation stabilizer, consistent with deficit zero in Theorem 2. At $N = 2$, every half-square is a nonzero scalar.

Finally let K have characteristic zero and contain a primitive Nth root ζ. Every matrix involved in the exact statements has entries in the cyclotomic subfield $\mathbb{Q}(\zeta)$. Matrix rank is unchanged under field extension, since it is characterized by vanishing and nonvanishing minors. An embedding of this subfield into $\mathbb{C}$ sends ζ to a primitive complex root. Such a root is ω_N^a with $\gcd(a,N) = 1$, giving the same minimax through a row-label permutation. The bit-reversal recurrence also holds directly for every primitive root. This proves the asserted field extension without assuming that all of K embeds into $\mathbb{C}$ or that $\sqrt{N} \in K$. It does not transfer the positivity proof to finite characteristic. This completes Theorem 1.

3 The finite abelian autocorrelation deficit theorem

Theorem 2. Let G be any finite abelian group of order $N \geq 2$, and let U be (3). Fix $R \subseteq \widehat{G}$ and $H \subseteq G$ with $1 \leq r = |R|, h = |H| \leq N$. Put $V = U[R, H]$ and

$$\begin{aligned} c_H(s) &= |H \cap (H - s)|, \\ d_H &= h - \max_{s \neq 0} c_H(s), \\ B_H &= \frac{h(N-r)}{N} - \frac{d_H r(N-r)}{N^2}. \end{aligned} \tag{17}$$

Then

$$\operatorname{rank} V \geq \left\lceil \frac{rh}{N} + \frac{d_H r(N-r)}{N^2} \right\rceil. \tag{18}$$

For every integer $0 \leq t < h - B_H$ and every $W \in \mathbb{C}^{r \times h}$ with $\operatorname{rank} W \leq t$,

$$\|V - W\|_2 \geq \sigma_{t+1}(V) \geq \left(1 - \sqrt{\frac{B_H}{h-t}}\right)^{1/2} > 0. \tag{19}$$

The transposed bounds hold with the roles of R and H interchanged. Explicitly, define d_R by nonzero translations in $\widehat{G}$ and put

$$B_R = \frac{r(N-h)}{N} - \frac{d_R h(N-h)}{N^2}.$$

Then the other rank bound is

$$\operatorname{rank} V \geq \left\lceil \frac{rh}{N} + \frac{d_R h(N-h)}{N^2} \right\rceil, \tag{20}$$

and the other gap is $(1 - \sqrt{B_R/(r-t)})^{1/2}$ when $0 \leq t < r - B_R$. One may take the larger rank lower bound and, among gap bounds whose own strict condition holds, the larger gap.

3.1 Proof of the rank estimate

Let $I = \widehat{G} \setminus R$ and $n = N - r$. Set

$$A = U[I, H]^* U[I, H], \qquad f_I(s) = \sum_{\chi \in I} \chi(s).$$

As in (6), A is a positive contraction and $V^*V = I_h - A$. For $x, y \in H$, $A_{xy} = N^{-1} f_I(y - x)$. Thus the exact second-moment identity and orthogonality give

$$\operatorname{tr}(A^2) = \frac{1}{N^2} \sum_{s \in G} c_H(s) |f_I(s)|^2, \quad \sum_{s \in G} |f_I(s)|^2 = Nn, \quad f_I(0) = n. \tag{21}$$

The identity uses $\operatorname{tr}(A^2) = \sum_{x,y} |A_{xy}|^2$, valid because A is Hermitian. Using $c_H(0) = h$ and $c_H(s) \le h - d_H$ for $s \neq 0$ yields

$$\begin{aligned} \operatorname{tr}(A^2) &\le \frac{hn^2 + (h - d_H)(Nn - n^2)}{N^2} \\ &= \frac{hn}{N} - \frac{d_H r n}{N^2} = B_H. \end{aligned} \tag{22}$$

The multiplicity of eigenvalue one of A equals $\dim \ker V$ and is at most $\operatorname{tr}(A^2)$. Hence $\operatorname{rank} V \ge h - B_H$. Taking the integer ceiling proves (18). Also $0 \le d_H \le h$, so $B_H = (n/N)(h - d_H r/N) \ge 0$, as required for (19).

3.2 Proof of the indexed gap and the transpose statement

List the eigenvalues of A as $a_1 \ge \cdots \ge a_h$, all in $[0, 1]$. The descending eigenvalues of V^*V are $1 - a_h, \ldots, 1 - a_1$. Therefore

$$\sigma_{t+1}(V)^2 = 1 - a_{h-t}.$$

Since at least $h - t$ eigenvalues of A are at least a_{h-t},

$$(h - t)a_{h-t}^2 \le \sum_{j=1}^{h} a_j^2 = \operatorname{tr}(A^2) \le B_H.$$

This proves the second inequality in (19). The strict condition $t < h - B_H$ implies $h - t > B_H \ge 0$, so the denominator is positive and the resulting gap is positive. This condition and (18) also ensure $t < \operatorname{rank} V \le \min(r, h)$, so the stated nonzero singular-value index is legitimate.

For completeness, let E be the span of the top $t + 1$ right singular vectors of V. Since $\operatorname{rank}(W|_E) \le t$, some unit vector $x \in E$ satisfies $Wx = 0$. Then $\|(V - W)x\|_2 = \|Vx\|_2 \ge \sigma_{t+1}(V)$, proving the first inequality in (19) without a separate approximation theorem as an unproved premise.

Under the canonical identification $G \cong \widehat{\widehat{G}}$, the transpose U^T is the unitary character table of $\widehat{G}$. Apply the proved statements to V^T, whose rank, nonzero singular values, and operator norm equal those of V. This proves (20) and the transposed gap. Theorem 2 is complete.

3.3 Endpoint and normalization cases

The definition of d_H does not apply to the trivial group, since there is no nonzero translation. At $N = 1$ the character table is [1]: its unique nonempty block has rank one and distance one from the only rank-zero matrix. Empty restrictions have rank zero. These direct facts are the entire $N = 1$ convention, not an artificial maximum over an empty set.

For $N \ge 2$, empty R or H may be handled directly as rank-zero blocks; they are excluded from the statement to avoid unnecessary deficit and approximation-index conventions. When $r = N$, $B_H = 0$ and V has orthonormal columns. Equations (18) and (19) give rank h and gap one for every $0 \le t < h$. When $h = N$, $d_H = 0$ and (18) gives rank r; the transposed bound gives the exact gap one for every $t < r$, since V has orthonormal rows.

For a singleton H, $d_H = 1$ and $B_H = (N - r)^2/N^2$. The rank bound rounds up to one and the gap at $t = 0$ is exactly $\sqrt{r/N}$, the norm of this column. For a singleton R, the transposed statement gives the

exact norm $\sqrt{h/N}$. These observations cover all nonempty restrictions at $N = 2$ as well as singleton cases at larger sizes.

If $t \geq h - B_H$, equation (19) asserts no strictly positive gap from that orientation. Equality at the threshold gives zero in the displayed expression; when $t \geq h$ the denominator is not used. Failure of this sufficient strict condition is not a claim that an exact rank-t representation exists. The other orientation may still apply.

For the unnormalized character table $F = \sqrt{N}\,U$, all rank conclusions are unchanged. For a rank-at-most-t approximant Z to $F[R, H]$, apply (19) to $W = Z/\sqrt{N}$: the lower bound on $\|F[R, H] - Z\|_2$ is $\sqrt{N}$ times the normalized bound. More generally, multiplication of the whole target by a scalar α multiplies the norm lower bound by $|\alpha|$. This is the normalization needed for every gap in this manuscript.

4 Existing consequences and their boundaries

Translation stabilizers and cardinality

For $s \neq 0$, $c_H(s) = h$ if and only if $H + s = H$. Thus $d_H > 0$ exactly when H has trivial translation stabilizer. If G is a p-group and $p \nmid h$, each nonidentity translation has orbit length a positive power of p, so H cannot be a union of its orbits. It follows that $d_H \geq 1$. More generally, $\gcd(h, N) = 1$ suffices: the nontrivial order of a translation divides N and would have to divide h for invariance.

For a square block of side s in a p-group with $p \nmid s$, (18) consequently gives

$$\operatorname{rank} U[R, H] \geq \left\lceil \frac{s^2}{N} + \frac{s(N - s)}{N^2} \right\rceil. \tag{23}$$

For example, at $N = 64$ and $s = 21$, the expression inside the ceiling is $29127/4096$, giving eight; the ordinary ceiling $\lceil 441/64 \rceil$ is seven. No location assumption or new separator model is needed for this existing example.

Critical rectangles for abelian two groups

For every finite abelian 2-group of order $N = 4L \geq 8$, take $r = 2L - 1$, $h = 2L + 1$. Odd cardinality gives $d_H \geq 1$ and (22) gives the same upper bound B_L in (8). With $t = L$, (19) yields the existing corollary

$$\sigma_{L+1}(U[R, H]) \geq \delta_N := \left(1 - \sqrt{\frac{B_L}{L + 1}}\right)^{1/2} > 0. \tag{24}$$

Indeed, $L < h - B_L$ is equivalent to $B_L < L + 1$. Since

$$L + 1 - B_L = \frac{1}{4} - \frac{1}{4L} - \frac{1}{16L^2} \longrightarrow \frac{1}{4},$$

the expansion $1 - \sqrt{1 - x} \sim x/2$ gives $\delta_N \sim (2N)^{-1/2}$. This is a decaying indexed gap, not uniform conditioning of the entire rectangle. If a full-matrix approximant has rank at most L on the corresponding critical corner, its operator-norm error from the independently reordered normalized character table is at least δ_N, since coordinate restriction cannot increase the error norm. A matching endpoint minimax upper construction for every noncyclic 2-group is not supplied.

Rank sums at the already accepted sizes

Let $S(M) = \sum_{k=1}^{N-1} \operatorname{rank} M[0 : k, k : N]$. Equation (16) gives the universal pointwise lower bound $\lceil k(N-k)/N \rceil$. For odd k and dyadic $N \geq 2$, the column size $h = N - k$ is odd, so (18) improves this to

$$\operatorname{rank} M[0 : k, k : N] \geq \left\lceil \frac{k(N-k)(N+1)}{N^2} \right\rceil. \tag{25}$$

At $N = 8$, these lower bounds are $(1, 2, 3, 2, 3, 2, 1)$. At $N = 16$, they are $(1, 2, 3, 3, 4, 4, 5, 4, 5, 4, 4, 3, 3, 2, 1)$. Recurrence (10) attains each entry. The exact minimum upper-cut sums at these sizes are therefore 14 and 48. The base-size minima are $S_1 = 0$, $S_2 = 1$, and $S_4 = 3$.

For the bit-reversal ordering only, summing (10) using reflection gives

$$S_N = \frac{N^2}{8} + \frac{N}{4} + 4S_{N/4}, \qquad S_1 = 0, S_2 = 1.$$

To see the coefficients, the even cuts up to the middle contribute $\sum_{a=1}^{L}(a + r_L(a))$, the odd cuts contribute $\sum_{a=0}^{L-1}(a+1+r_L(a))$, and reflection doubles the combined sum and subtracts the middle value $r_N(2L) = L$. For $N = 2^m$, the solution is

$$S_N = \begin{cases} N^2/6 + N(m/8 - 1/6), & m \text{ even}, \\ N^2/6 + N(m/8 + 1/24), & m \text{ odd}. \end{cases} \tag{26}$$

Meanwhile summing the unrounded bound in (16) gives $S(M) \geq (N^2 - 1)/6$. Thus (26) has the optimal leading term $N^2/6$. It does not prove exact all-size minimum rank sums. Already at $N = 32$, (25) together with the even-cut baseline sums to 182, whereas (26) gives 192; these bounds do not settle the gap.

5 Prior art and the recovered final paper comparison

Double digit reversal and the simultaneous rank-one block hierarchy are established in Delvaux and Van Barel's final 2008 article [1], especially Eq. (15) and Lemma 7. The upper calculation in Section 2 is explicit cut bookkeeping within that known structure. Their Theorem 9 describes the minimum Fourier support of one nonzero vector with bounded input support; Remark 10 gives additional extremal-support geometry. The $N = 8$ critical lower bound is already a consequence: a rank-deficient 5×3 transposed critical block would give input support at most three and output support at most three, whereas their theorem requires at least four.

For $N = 4L \geq 16$, the direct one-vector substitution at a hypothetical rank L only gives Fourier support at least four, while the two complement sizes are $2L - 1$ and $2L + 1$. This direct substitution does not exclude rank L. That observation distinguishes quantifiers; it is not a proof that the entire older theory cannot imply the general result through a different argument.

The final comparison report [S6] has now been recovered byte-for-byte at its recorded hash, and the local final journal PDF matches the hash recorded in that closure. The relevant final passages were checked again against this PDF. Its source-availability issue is closed. Older immutable reports [S2, S5] contain superseded access wording; that wording is not a current blocker. The bounded comparison concluded that the complete general minimax and deficit rank/gap statements were not stated in that article. It did not establish worldwide novelty.

The ordinary flat-unitary bound (16) is classical; de Wolf [4, Theorem 1] gives the Hadamard version, attributes it to Lokam, and discusses the equal-magnitude orthogonal/DFT setting. Projection and Hilbert–Schmidt uncertainty methods are classical [7]. Stabilizer-sensitive support uncertainty is also established [5, Lemma 7.1 and Theorem 7.2]. The statistic used here is not newly invented: Shkredov [6, Eq. (4)] uses $\rho(H) = \max_{s \neq 0} |H \cap (H + s)|$, and exactly $d_H = h - \rho(H)$. The sign of the translation has no effect on this maximum.

The Kronecker Fourier work [3] is relevant prior art for subgroup-based rank-deficient blocks and support extrema on general finite abelian groups. Merely extending the group class is not an originality argument. The remaining historical comparison concerns the exact overlap-retaining fixed-rectangle coefficient, higher-dimensional support or rank profiles, and concentration second moments. Those questions are separate from the complete proofs above.

Versioned citation concordance

Source and version	Hierarchy or support passages	Verification in this packet
DV2008 final [1]	Eq. (15), pp. 1593–1594; Lemma 7, p. 1596; Theorem 9, pp. 1598–1599; Remark 10, p. 1599	Recovered accepted comparison; matching local final PDF; key passages rechecked
TW470 September 2006 [2]	Lemma 8, printed p. 11 / PDF p. 13; Theorem 10, printed pp. 13–14 / PDF pp. 15–16	Author-institution PDF and numbering freshly checked
TW477 November 2006 [3]	Theorem 14, printed pp. 11–12; Theorem 17, p. 13; Theorem 20 and Corollary 21, pp. 17–18	Author-institution preprint; earlier bounded comparison retained; no relabeling as final-journal numbering

The preprint and final article numbers are deliberately not interchangeable. The local inspection did not repeat the entire final-paper comparison or all earlier specialist literature searches. References [4]–[7] were checked at their indicated primary-source versions; the broader inherited reference inventory is preserved in the source packet.

6 Status and open extensions

No mathematical gap was found in consolidating the two stated theorems. Their previously recorded independent checks are [S3] and [S4]. The present manuscript is an author-side consolidation awaiting a separate independent manuscript review; it does not label itself newly independently accepted. The algebraic controls accompanying the packet are regression checks, not replacements for the proofs or new finite-field theorems.

The following are optional future problems, not omitted assumptions or unfinished steps in either proof:

- Find matching endpoint minimax upper constructions for arbitrary noncyclic finite abelian 2-groups.
- Determine the exact minimum upper-cut rank sum at every dyadic size beyond the accepted small cases.
- Determine sharper or optimal fixed-rectangle gap bounds, or suitable distinct positive-characteristic rank statements.
- Complete the wider model-matched historical comparison for the general minimax and the exact deficit formula.

No separator fusion is developed in this manuscript. Nor does high cut rank alone measure Fourier evaluation cost: even a permutation can have large corner rank while applying cheaply. Arithmetic or circuit lower bounds would require a separately specified and charged computation model. Neither theorem supplies one.

References

1. Steven Delvaux and Marc Van Barel. *Rank-deficient submatrices of Fourier matrices.* Linear Algebra and its Applications 429 (2008), 1587–1605. doi:10.1016/j.laa.2008.04.043. Final journal version inspected locally; source PDF SHA-256 is recorded in the packet.
2. Steven Delvaux and Marc Van Barel. *Rank-deficient submatrices of Fourier matrices.* Report TW470, September 2006. Author-institution preprint. Numbering is specific to this version.
3. Steven Delvaux and Marc Van Barel. *Rank-deficient submatrices of Kronecker products of Fourier matrices.* Report TW477, November 2006. Author-institution preprint. The later article is in Linear Algebra and its Applications 426 (2007), 349–367; passage references here use TW477.
4. Ronald de Wolf. *Lower Bounds on Matrix Rigidity via a Quantum Argument.* Author-hosted preprint, Theorem 1, p. 4, and Section 1.2, p. 2. The flat-matrix lower bound is credited there to Lokam; no unverified original corollary number is asserted here.
5. Tao Feng, Henk D. L. Hollmann, and Qing Xiang. *The shift bound for abelian codes and generalizations of the Donoho–Stark uncertainty principle.* arXiv:1804.00367v1, 2018, Lemma 7.1 and Theorem 7.2, PDF p. 11.
6. Ilya D. Shkredov. *Uncertainty for convolutions of sets.* arXiv:2404.12469v1, 2024, Eq. (4), p. 2. The cited definition is the maximum nonzero-translation overlap.
7. Gorjan Alagic and Alexander Russell. *Uncertainty Principles for Compact Groups.* arXiv:math/0608702v2, 2008, Theorem 1 and Corollary 1, pp. 1–2, and Theorem 3.1.1, p. 4.

Appendix Proof source concordance

The following labels refer to exact local source copies in the accompanying proof-source packet. Complete byte counts, original paths, retrieval provenance, and full SHA-256 hashes are in SOURCE_MANIFEST.json. The source copies remain unchanged, including their historical dating and now-superseded access prose.

Label	Source	Role in this manuscript
S1	MINIMAX_PROOF, *Independent Fourier endpoint rank audit*	Theorem 1, exact recurrence, small rank sums
S2	DEFICIT_PROOF, *Autocorrelation-deficit Fourier bounds and routed separators*	Theorem 2, stabilizer/cardinality consequences; separator sections excluded here
S3	R07, REVIEW-07 report, Section 7 C02	Recorded independent acceptance of endpoint scope and critical rectangle extension
S4	R08, REVIEW-08 report, deficit passage	Recorded independent checking of the separate deficit estimate and spectral index
S5	MINIMAX_PRIORITY, endpoint priority report	Half-square boundary, prior-art boundaries, version-specific references
S6	Recovered DELVAUX_FINAL_COMPARISON	Half-square proof and completed bounded final-paper comparison

Label	Source	Role in this manuscript
S7	FINAL_THEOREMS, Section 1	Existing abelian two-group critical-rectangle gap and detailed upper proof
S8	ORIGINAL_MINIMAX_PROOF	Original exact recurrence exposition extracted inertly from its uploaded archive

The five readiness-pinned main sources were rehashed and matched before writing. Source S6 was recovered from the repository blob already listed in the local pinned tree; it matches its inherited comparison hash exactly. The packet does not redistribute the journal article PDF or its extracted full text. It records the verified local PDF hash and its bibliographic reference instead.